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TS and TM - Batch File Upload Instructions (SFTP or API)

2 months ago · Updated

Flat File Uploading - Introduction

Clients who have not contracted to send real-time data (transactions or trade/market data) to HALO will upload .csv files to HALO either via a secure SFTP drop point (for trade data only) or via the File Uploader service API (transactions or trade data) - both of which involves the files being uploaded to a private location (an AWS S3 bucket) which are then picked up for processing by HALO Trade/Market Surveillance and Transaction Monitoring module algorithms.

As a best practice approach, we suggest uploading a small representative sample of the file (small data set representing the common trades and asset types). Once the file is processed, you can verify it in HALO Data Audit tab, and if there are no errors found, you may upload a larger full data set.

1. SFTP (Trade data only)

a. Provide Solidus Labs your IP/s and Public SSH key handshake

The first step in setting up the SFTP drop point is to provide your Solidus Labs technical contact the IP/s from which the files are expected to be uploaded from at your institution as well as your public SSH key.

b. Solidus Labs provides you the SFTP connection information

With the above information, Solidus Labs will produce a unique secure and private SFTP drop point in the format: `yourinstitution@sftp-<HALO-DOMAIN>`

c. Ready to connect and send data files

From your institution's IP/s, upload the file/s to the SFTP drop point in the agreed upon cadence.

Note: One SFTP endpoint will be made available for your test UAT instance and one for your Production instance.

File Guidelines: Naming convention, Sizes and Limitations:

- File naming convention: YourOrgFileEntityType_DateDDMMYYYY.csv
 - FileEntityType = order or execution



yourorgname_ExecutionData_YYYYMMDD_part1.csv
 yourorgname_OrderData_YYYYMMDD_part1.csv

- Each .csv file must contain the file column headers of the fields being sent
- No periods or special characters - other than underscore - should precede the .csv suffix of the filename(s)
- The maximum allowed single file size is 500 MB. Uploads larger than 500 MB may fail, so files must be split if the total size exceeds 500 MB.
 - If splitting files, ensure the file column headers exist in each split file.
- If you plan to send more than one file daily, append a suffix such as: fileType_DDMMYYYY_part1 (and _part2, etc.).
- If sending multiple files, daily processing will be more efficient if each file is roughly the same size (ideally 50–300 MB), rather than sending many small files.

Note: for **historical data uploads** (over 7 days old) please use the following naming convention to route the file into the right place:

- For executions, the filename should include the string "historical_execution"
- For orders, the filename should include "historical_order"

2. File Upload to HALO

For the URLs noted, you must replace **<HALO-DOMAIN>** with one of the following instances, as noted [here](#). Please ask your Solidus Labs Technical Contact to confirm your environment name if there is any doubt.

Steps to upload data to HALO:

- **Authenticate/Receive API Security Token**

See [authenticating for API and receiving your security token](#)

- **Request a file upload link**

- You need to request a one-time-use-only link to upload the file. To upload another file, please call the endpoint with the security token received in the previous step to get the upload link. The file upload link is valid for 15 minutes only.

The following cURL command will request the file-uploader to issue an upload link to your private storage at Solidus Labs.

POST- <https://file-uploader.<HALO-DOMAIN>/issue-upload-url>

```
curl --request POST 'https://file-uploader.<HALO-DOMAIN>/issue-upload-url' \
--header 'X-Auth-Token: your-security-token-here' \
--header 'Content-Type: application/json' \
--data-raw '{ "fileName": "your_file_name.csv", "fileType": "file_type" }'
```

- **Supported FileTypes:**
 - For **Trade/Market Surveillance**, these are:
 - Order File Types
 - *PRIVATE_ORDER (v1 Schema) (Legacy)*
 - PRIVATE_ORDER_V2 (v2 and v2.1 Schema)
 - Execution File Types
 - *PRIVATE_EXECUTION (v1 Schema) (Legacy)*
 - *PRIVATE_EXECUTION_V2 (v2 Schema) (Legacy)*
 - LINKED_PRIVATE_EXECUTION_V2 (v2.1 Schema)
 - For **Transactions**, the supported FileType is
 - TRANSACTION - transaction files

- **File name format:** fileType_DateDDMMYYYY.csv

If you plan to send more than one file (see [file sizes/limitations below](#)), then you can append it like



- **Sample response:**

```
{
  "dropLocationUrl": "https://solidus-file-watcher-staging.s3.amazonaws.com/solidusClient%3D***/fileType%3Dprivate_order/date%3DYYYY-MM-DD/your_file_name.csv_YYYYMMDD?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Date=YYYYMMDDTHHMMSSZ&X-Amz-SignedHeaders=host&X-Amz-Expires=900&X-Amz-Credential=***&X-Amz-Signature=***",
  "message": "ok"
}
```

- **Upload the file(s)**

Using the link you've received via the previous command and path of the file you'd like to upload, please call the following endpoint to upload the file.

```
curl --request PUT your-aws-upload-link \
--header 'Content-Type: application/octet-stream' \
--upload-file '/path/to/your/file/here'
```

Sample response:

```
Status: 200 OK
Time: 116ms
Size: 261B
```

- **Verification**

- The following endpoint shows you the list of the files you have sent.

- GET- <https://file-uploader.<HALO-DOMAIN>/list-files>

```
curl --request GET 'https://file-uploader.<HALO-DOMAIN>/list-files' \
--header 'Content-Type: application/json' \
--header 'X-Auth-Token: your-security-token-here'
```

Sample response:

```
{
  "msg": "ok",
  "entries": [
    {
      "filePath": "/processed/private_order/your_file_name.csv_YYYYMMDDHHMMSS",
      "lastModified": 1644010167.048633000
    },
    {
      "filePath": "/processed/private_order/your_file_name2.csv_YYYYMMDDHHMMSS",
      "lastModified": 1644009940.985685000
    },
    {
      "filePath": "/processed/private_order/your_file_name3.csv_YYYYMMDDHHMMSS",
      "lastModified": 1644008417.202175000
    },
    {
      "filePath": "/processed/private_order/your_file_name4.csv_YYYYMMDDHHMMSS",
      "lastModified": 1644008416.640827000
    },
    {
      "filePath": "/processed/private_order/your_file_name5.csv_YYYYMMDDHHMMSS",
      "lastModified": 1644005590.980308000
    },
    {
      "filePath": "/processed/transaction/your_file_name6.csv_YYYYMMDDHHMMSS",
      "lastModified": 1643921260.336390000
    },
    {
      "filePath": "/processed/private_execution/your_file_name7.csv_YYYYMMDDHHMMSS",
      "lastModified": 1643921260.095714000
    },
    {
      "filePath": "/processed/broker_execution/your_file_name8.csv_YYYYMMDDHHMMSS",
      "lastModified": 1643921259.894958000
    },
    {
      "filePath": "/processed/private_order/your_file_name9.csv_YYYYMMDDHHMMSS",
      "lastModified": 1643921258.989392000
    }
  ]
}
```

3. Both SFTP and File Uploader: Verify File processing in HALO

The **Data Audit** tab in HALO Dashboard will show the number of records in 'Processed' vs. 'Errors' based on Data Ingestion which is the first processing phase. This ingestion process runs every hour. The algo batch processing runs once a day as per the batch job times given above.



Within the last						
Day	Enter a date range		Search			
	11/28/2022 – 11/28/2022					
Date	File Name	Incoming Records	Record Type	Processed	Errors	Duplicates
Nov 28, 2022, 4:30	transaction_27112022.csv	8148	Transactions	8148	—	—

File Sizes / Limitations

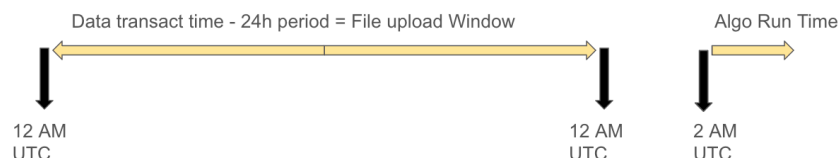
The maximum allowed file size is 500 MB and the process will fail if a file larger than this is uploaded. Splitting files is required when the total file size would be larger than 500MB. When sending multiple files, the system will be optimized if each file is sent approximately the same size (between 50MB-300MB as opposed to many small files). Please ensure that each split file contains the File header described above.

Batch Processing Times:

Clients may send one file/day or multiple files throughout the day.

Alerts are generated T+1 after the transaction date of the ingested files. The batch process time to consume the files received is configurable based on the client's requirements, as noted below. The Algos will run once a day regardless of how many files are sent throughout the day.

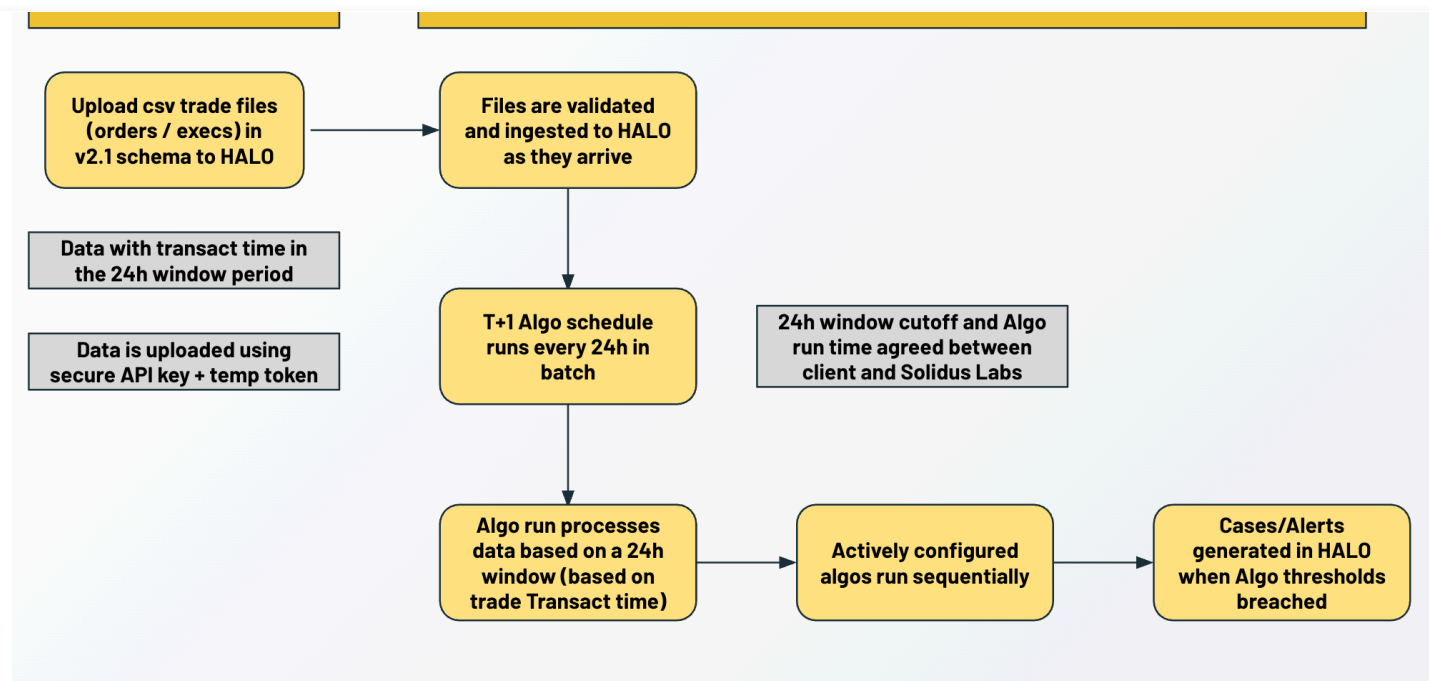
An example of a File Upload Timeline is shown below:



The points to highlight in the example above are as follows:

- File upload window and Cut-off Time (aka data-fencing window):** This is the allowed period when we can receive data related to the *data window*. Clients are free to define this *upload cut-off time* to accommodate their needs. **Data window:** defines the transaction time window for the trading data to be processed in each algo run. It is always a 24-hour period, and the default is 12 am UTC to 12 am UTC the next day. However, the customer can choose a different 24-hour period (e.g. 7 am UTC to 7 am UTC) to align to their local 12 am-12 am time window. The file upload window is simply a buffer to ensure we have plenty of time to ingest, normalize and enrich the data received before the Algos process it.
- Algo Batch run time:** The time at which the data received for the last *data window* will be processed by our Algo Batch Process. This is normally scheduled to run 2 hours after the *upload cut-off time* to accommodate large amounts of data.

Trade / Market Surveillance File-Based Batch Workflow



Note: The TS Algo batch process is designed to process incoming order and execution data, where transact-time is within the last 24hr period only. Anything older than that will not be processed.

Timings to consider here:

1. Client **data Sending Time/Upload Time** (actual time that you send data)
2. Client **data Transact time** of given data (the time the transaction took place on an exchange)
3. **File ingestion time** period and frequency by HALO in which to process incoming data
4. **Batch Algo runtime** on processed data
5. **Alert generation time** and display in HALO

Additional Notes:

- If the data for the defined 24hr period is not received by the cut-off time period and is sent later or on the next day, it won't be processed in the next batch automatically. This is because the transaction time is older than the defined 24 hrs for the given day.
- If this remaining data needs to be processed, it will be done as part of a [historic data run](#).
- ADV for a day may be affected by incomplete data, and you may have missing alerts.
- If the complete data set is resubmitted and processed as part of a historic run, you may see duplicate alerts.
- The standard Algo batch runs for the last 24hr period however, if it's needed to be run for an older period, then it's a manual process that is to be planned with the technical team.
- If the data is older than 7 days, then it should be sent to the historic data endpoint, and the effort is to be planned with your CSM.

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